
Lecture Notes

on Combinations, Permutations, and Binomial Coefficients

by Arnaja Mitra

1 Permutation, Combination and Binomial Coefficient

1.1 Factorial

$$n \in \mathbb{N} = \mathbb{N} \cup \{0\}$$

The **factorial** of a number n (where $n \geq 1$, integer) is defined as

$$n! = n \times \underbrace{(n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1}_{(n-1)!} = n(n-1)!$$

Examples:

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120, \quad 1! = 1.$$

Note: $0! = 1$

1.2 Permutation

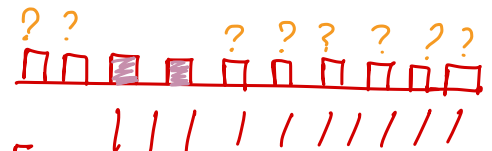
A **permutation** is an arrangement of objects in a particular way or order. Order is essential in permutation.

We denote by $P(n, k)$ or ${}^n P_k$ a permutation of selecting k objects from n objects, and it is defined as

$$P(n, k) = \frac{n!}{(n-k)!}, \quad n \geq k.$$

This expression gives the number of permutations of k objects selected from n objects where order matters.

Examples on Permutations



Recall:

$$P(E) = \frac{\text{Number of favorable outcomes in } E}{\text{Total number of possible outcomes}}$$

☑ **Example 1:** How many ways can the 3rd and 4th positions be awarded to 10 members?

Here $n = 10$, $k = 2$.

$$\begin{aligned} \therefore \text{ways of choosing 3rd and 4th position among 10 members} \\ = P(10, 2) = \frac{10!}{(10-2)!} = \frac{10 \times 9!}{8!} = \frac{10 \times 9 \times \cancel{8!}}{\cancel{8!}} \\ = 90. \end{aligned}$$

☑ **Example 2:** How many 3-letter words (with or without meaning) can be formed from the letters of the word **SWING** when repetition of letters is not allowed?

$n = 5$, $k = 3$, so the required number of ways

$$= P(5, 3) = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times \cancel{2!}}{\cancel{2!}} = 60.$$

1.2.1 Permutation When Repetition is Allowed

Say the total number of objects = n , and you want to select r among them, when **repetition is allowed** (i.e., you can select the same object more than once).

Each time, an object can be selected in n ways, and you make r selections.

Total number of ways = $n \times n \times n \times \cdots \times n$ (r times).

$\therefore$ Total number of permutations **when repetition is allowed** = n^r .

✎ **Example:** How many 3-letter words (with or without meaning) can be formed out of the letters of the word **SWING** **when repetition of letters** is allowed?

$5 \times 5 \times 5$

Number of ways = $5^3 = 125$.

Explanation:

- The first letter can be chosen in 5 ways.
- Since repetition is allowed, the second letter can again be chosen in 5 ways.
- Similarly, the third letter can also be chosen in 5 ways.

$\therefore$ Total number of arrangements = $5 \times 5 \times 5 = 125$.

1.3 Combination

A **combination** is defined as an arrangement of objects where the order in which the objects are selected **does not matter**.

Order is not important in combination.

We denote by $C(n, k)$ or nC_k a combination of selecting k objects from n objects, and it is defined as

$$C(n, k) = \frac{P(n, k)}{k!} = \frac{n!}{(n-k)! k!}, \quad n \geq k.$$

This expression gives the number of combinations of k objects selected from n objects.

Examples on Combination

Example 1: Find the number of subsets of the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ having 3 elements.

$$S = \{1, 2, 3, \dots, 10\}, \quad E \subseteq S \text{ such that } n(E) = 3.$$

$$E = \{1, 2, 3\}, \{2, 1, 3\}, \{3, 1, 2\},$$

(Note that order does not matter in combinations.)

*→ same when combination
→ different when permutation*

Here $n = 10, k = 3$.
 $\therefore$ Numbers of ways to have 3 elements set E is given by

$$C(10, 3) = \frac{10!}{(10-3)! 3!} = \frac{10 \times 9 \times 8 \times \cancel{7!}}{\cancel{7!} 3!} = \frac{10 \times 9 \times 8}{3 \times 2} = 120.$$

Example 2: A group of 3 lawn tennis players S, T, U . A team consisting of 2 players is to be formed. In how many ways can we do so?

Here we have, $n = 3, r = 2$.
 The required numbers of total ways to form a 2 players team

$$= C(3, 2) = \frac{3!}{(3-2)! 2!} = \frac{3 \times \cancel{2!}}{1! \cancel{2!}} = 3.$$

Note: The quantity nC_r or $C(n, r)$ is known as the **binomial coefficient**, and is also written as

$$\binom{n}{r} = \frac{n!}{(n-r)! r!}.$$

Using Binomial Coefficients in Probability

1.4 Binomial Probability Formula

If an experiment consists of n independent trials, and each trial has only two possible outcomes: “**success**” with probability p , and “**failure**” with probability $q = 1 - p$, then the probability of getting exactly r successes in n trials is given by the **Binomial Distribution**:

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \quad r = 0, 1, 2, \dots, n.$$

Here,

- $\binom{n}{r}$ counts the number of ways r successes can occur in n trials,
- p^r is the probability of those r successes, and
- q^{n-r} is the probability of the remaining failures.

$$\begin{array}{cc} q & p \\ \downarrow & \downarrow \\ k_1 & k_2 \\ k_1 + k_2 = 1 \\ \Rightarrow k_2 = 1 - k_1 \\ \hline p + q = 1 \end{array}$$

Tossing 1 time
 $S = \{H, T\}$

Example on Binomial Probability

Example: A fair coin is tossed 5 times. Find the probability of getting exactly 3 heads.

Here $n = 5$, $p = \frac{1}{2}$. Now, let H denote the event of getting head & T " " " " " tail.

$$p = P(H) = \frac{1}{2}. \quad \therefore q = 1 - p = 1 - \frac{1}{2} = \frac{1}{2}. \quad \binom{n}{r} p^r q^{n-r}$$

Thus the probability of getting exactly 3 heads = $\binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^{5-3}$

$$= \frac{5!}{(5-3)!3!} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \frac{5!}{2!3!} \left(\frac{1}{2}\right)^5 = \frac{5 \times 4 \times 3!}{2!3!} \frac{1}{32}$$

$$= \frac{5 \times 4}{2 \times 1} \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

1.5 Summary Table

Symbol	Meaning
n	Number of trials (repetitions)
r	Number of successes
p	Probability of success in one trial
$q = 1 - p$	Probability of failure in one trial
$P(X = r)$	Probability of exactly r successes in n trials

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

$$P(X = r) = \binom{n}{r} p^r q^{n-r}$$

$$a^s a^t = a^{s+t}$$

$$2^2 2^5 = 2^7$$

$$\left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^5$$